

Approve the Rule, Not the Case

Separating how much oversight a decision needs from what the decision is

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Abstract

Putting a human in the loop on every consequential decision does not scale, and what it degrades into is rubber-stamping. Removing the human leaves nobody accountable. Most designs choose a point on that line and live with it.

We argue the dilemma comes from conflating two different questions. *How much oversight does this decision need?* is a policy over consequences: harm, reversibility, whether a right is engaged, how complete the evidence is. *What is the decision?* is logic. Answering both with the same mechanism forces the trade-off; answering them separately dissolves it.

We built a decision console on that separation. An oversight ladder allocates one of four control tiers and never touches the answer. A logic database of approved rules derives the answer and never decides how much supervision it needs. Between them a gate asks one question: *does an approved rule cover this case?* If so the decision proceeds, whatever its tier, because a rule an expert approved *is* their decision and applying it executes a call they have already made. If not, and the consequences warrant it, the inquiry waits for a person.

The system runs. Over a twelve-case load starting from an empty database, 18–30 per cent of decisions were made deterministically from approved logic. We report what broke in building it, and we report why that figure is not higher: nothing generalises. The database accumulates one clause per case *shape*, and the evidence is legible in its own vocabulary, which holds `reorder_toner` and `reorder_paper` side by side where `reorder(C, X)` would do.

1 The problem with a human in the loop

Oversight of automated decisions is usually specified per decision. A person reviews each consequential output before it takes effect, and the arrangement is called meaningful human control.

It fails in two directions at once. At volume, per-decision review is unaffordable, so it is either skipped or performed at a rate that makes it ceremonial. And a reviewer shown a recommendation without the reasoning that produced it cannot do much besides agree: the check that was supposed to catch errors becomes a signature that launders them.

The response is usually to tune the threshold — review the top few per cent, sample, escalate on low confidence. That manages the cost without addressing the cause, which is that the same mechanism is being asked to answer two unrelated questions.

2 Two halves

The first question is *how much oversight does this decision warrant?* It is answered from properties of the decision, not of the answer: how badly a wrong call hurts the person it lands on, how readily it can be undone, whether a right is engaged, how complete the

evidence is. That is a policy. It should be written down, argued over, and applied the same way on Tuesday as on Monday.

The second question is *what is the decision?* That is logic — conditions and conclusions — and it should be derived, inspectable, and identical for identical inputs.

Our system answers them with two mechanisms that do not share state beyond a single boolean. The oversight ladder produces a tier and cannot see the conclusion. The logic database produces a conclusion and cannot see the tier. Between them sits a gate.

3 Allocating oversight

The ladder is not ours. It is the harm/reversibility/rights test from our group’s earlier prototype, itself a restatement of the structure running through the EU AI Act and the assurance literature. Four inputs, four tiers, evaluated top down:

```
if (harm >= 9 and rights) or (harm >= 8 and reversibility <= 3):
    human_only      # AI may not decide
if harm >= 6 or rights or confidence < 70:
    human_approval  # a person approves before it takes effect
if harm <= 3 and reversibility >= 7:
    ai_with_oversight # bounded autonomy, logged and reversible
ai_recommends      # analyse and propose; do not act
```

Nine lines, in a module that imports nothing — not the graph framework, not the model client, not the rest of the package. A module that cannot reach a network cannot be talked into anything, and this is the one that decides how much supervision a decision gets.

The ladder also exists a second time, in JavaScript, so the console’s sliders respond without a round trip. Two copies of the rule allocating human oversight is a hazard: they can drift with nothing failing, and the failure mode is a page showing one tier while the server acts on another. A test extracts *classify from the shipped page*, runs it under Node, and compares all 1,000 input combinations against the Python. We verified it bites by mutating one threshold; it reported twenty disagreements and named them.

3.1 What a model may contribute here

A language model estimates the four inputs from the conversation and returns nothing else. It is told explicitly that it does not decide what oversight follows from its numbers. The separation is not decoration: estimating harm is a judgement about the world, which a model may reasonably attempt; deciding what supervision follows is a policy, which must be stable.

Both the model’s rubric and the help text behind each control on the page are generated from one source. If the person moving a slider and the model estimating it work from different definitions of harm, the two numbers look comparable and are not.

Table 1: What the gate does. The tier sets what is at stake, not whether a person is asked.

Tier	Approved rule fires	Nothing fires
human_only	proceeds	waits
human_approval	proceeds	waits
ai_with_oversight	proceeds	proceeds, sampled
ai_recommends	proceeds	proceeds, sampled

4 Deciding the case

The decision itself comes from function-free Horn clauses with negated body literals — Datalog — evaluated by forward chaining to a fixpoint. No function symbols, so the derivable set is finite and termination is structural rather than a timeout. Each round matches a snapshot, so a rule’s result cannot depend on where in the round it ran. Forward rather than backward, so the whole derivation exists when it finishes and the explanation is a transcript of what happened rather than an account reconstructed afterwards.

Facts are three-valued, and the third value carries the weight. Facts arrive from a model reading free text; if it did not mention a formal notice, that does not mean none was given — it means nobody said. Under a closed world that silence becomes false, the negation succeeds, and a rule fires against a person on evidence nobody gathered.

```
| will_get_wet(C) ← going_out(C), raining(C),
|                  not has_umbrella(C).
```

With the umbrella recorded absent, the rule fires. With the umbrella *unmentioned*, it does not, and the head is recorded as blocked rather than false, so anything reading `not will_get_wet` sees UNKNOWN too. Ignorance has to survive negation, or it becomes evidence one round later. We shipped that defect twice in the predecessor system and reproduced it once here before the tests caught it.

5 What the model may do

The agent is a small graph. Its shape encodes what a model is permitted to contribute:

```
intake → suggest → classify → extract → derive → propose →
gate → reply
```

Four of those call a model. `intake` decides whether a decision has been described at all. `suggest` estimates the four oversight inputs. `extract` rewrites the conversation as ground facts. `propose` drafts candidate rules. None of them produces an answer.

The three that do not call a model are the ones that matter. `classify` runs the ladder. `derive` runs the engine. `gate` decides whether a person is needed. Same inputs, same outputs, every time.

6 The gate

The gate asks one question: *does an approved rule cover this case?*

That approved logic proceeds even at `human_only` is the design’s most contestable claim, and it is deliberate. A rule an expert approved is their decision: they saw the clause, they accepted that any case matching its body gets its conclusion, and applying it executes a call they have made rather than making a new one. Requiring them to confirm it again for every matching case is how oversight becomes ceremony — the failure this design exists to avoid.

Two things keep that defensible. Approval is of a *clause*, whose body states exactly which cases it claims; and a proportion of decisions is audited afterwards regardless, at a rate that is highest where a wrong decision hurts most — 50 per cent for `human_only` logic, 5 per cent for routine — and never zero, because a rule right a thousand times can start being wrong when the population shifts underneath it. Sampling is deterministic on a hash of the case, so an auditor can recompute why any particular case was or was not checked.

A parked inquiry lives in the database, not in the framework’s checkpoint. An in-memory checkpoint loses every pending approval on restart, leaving the person who was waiting waiting for something that no longer exists. Nothing is said to them in the meantime: a holding answer gesturing at the likely outcome would *be* the decision, made without the person whose job it was.

7 What broke

The system was built against a live model throughout. Five failures are worth reporting, because four of them would have shipped silently.

A greeting was classified. Typing “Hi” produced *human approval required*. The estimator, correctly, scored confidence 0 and wrote “no decision has been described yet” — and the ladder, also correctly, read confidence < 70 and demanded a person. The architecture forced a classification out of a conversation. The fix was a separate intake phase that classifies nothing.

The intake gate overfitted its own prompt. We first judged it flaky. Measured, it was deterministically wrong: 0/6 handover on one ordinary case and 6/6 on a near-identical one, the only difference being that the second appeared in the prompt’s example list. Rewriting the examples as an applicable test made it worse. Four rounds of prompt wording with inconsistent results was the signal to change shape rather than words: the gate now asks the *narrow* question — was this purely conversational? — and hands over unless the answer is yes. Recognising a greeting is something the model does reliably; recognising a decision is not. It fails open, because being wrong that way costs a moment of needless assessment while being wrong the other way means a decision is never assessed at all.

Extraction returned prose. We had folded fact extraction into the estimator’s call to save a request. One system prompt doing two jobs blended them: the model stayed in assessment voice and returned “Tenant disputes arrears amount” where `debt_disputed(c)` was required. Every fact was dropped at parse time, so the rule base could never fire — cheaper, and useless. Extraction now has its own node and its own prompt.

The proposer drafted a rule for the opposite case. Given a disputed debt with no notice given, it produced a clause requiring `not debt_disputed(c)` and `formal_notice_given(c)`: logic that could never fire on the case that prompted it. Asking more firmly is not a fix. Every proposal is now run against its own case before it reaches the review queue and discarded if it does not fire.

Parked inquiries contributed nothing. The graph routed a waiting case straight to its end, bypassing abstraction — so the most consequential decisions in the system, the ones a person had to

Table 2: Two runs of the same twelve-case load from an empty database.

	Run A	Run B
Decided from approved logic	2 / 12	3 / 12
Parked for a person	5 / 12	4 / 12
Coverage	18%	30%
Approved rules	7	5
Distinct predicates coined	19	15

look at, were exactly the ones whose logic was never written down. Abstraction now runs before the gate.

8 Coverage, and why it is low

The measure that matters is the share of decisions made from approved logic rather than from a model. It starts at zero and every approved rule should move it; a plateau means the database has become a cache of past answers rather than a body of logic.

We ran a twelve-case load past an empty database twice, approving every rule the agent proposed. Approving generously is deliberate: the question is whether approved logic gets *reused*, not whether a reviewer would have accepted it.

The mechanism works: rules are proposed, approved, and reused, and coverage climbs within a run. What it does not do is climb far, and the reason is legible in the vocabulary the system built for itself. Run B coined, among fifteen terms:

```
| reorder_toner      reorder_paper
| toner_stock_low   paper_stock_low
| toner_stock_depleted paper_stock_depleted
```

Six terms and two rules for one idea. A case about toner produces logic that cannot decide a case about paper, so the second case arrives uncovered and produces its own near-duplicate. Where the extractor happened *not* to bake the entity into the predicate — *in arrears*, *debt disputed*, *formal notice given* — coverage appeared immediately, and three tenancy cases were decided by logic drawn from the first.

Coverage happens exactly where the terms do not carry the entity, and fails exactly where they do. The engine is first-order and can already express $\text{reorder}(C, X) \leftarrow \text{stock_low}(C, X)$, not $\text{depleted}(C, X)$. Nothing proposes it.

9 Limitations

Nothing generalises. This is the finding, not an aside. Without a step that merges near-duplicate rules into a variable-carrying general one, coverage creeps rather than climbs, and a system that does not generalise offers no determinism in practice however deterministic its engine. The missing mechanism is anti-unification over approved rules, replayed against the stored cases so a reviewer sees where the generalisation would have disagreed before approving it. The case log exists for exactly that and is not yet used for it.

No expert has used this. Every approval in Table 2 was granted by a script. Whether a reviewer can judge a proposed clause, and how long it takes them, is the study we have not run and the assumption everything rests on.

One model, one size. All measurements are Claude Haiku 4.5. Whether a stronger model coins a more reusable vocabulary — which is the whole bottleneck — is untested and cheap to test.

Conduct at the top tier is a prompt, not a mechanism. When a case does reach *human_only*, the reply is constrained by an instruction not to recommend an outcome. It held across every trial we ran, including direct pressure to give a one-word answer, but three trials is an anecdote. Nothing inspects the reply before it reaches the person.

Two runs is not a measurement. The spread between 18 and 30 per cent on identical input is wide, and it is driven by whether the extractor happened to reuse a term. We report the range rather than the better number.

10 Related work and positioning

Datalog evaluation, stratified negation and canonical clause forms are standard; we implemented rather than imported them because the system needs three-valued facts and proof retention, and because the properties being claimed should be ones we can point at in the source.

The contribution is not the engine. It is the observation that oversight allocation and decision derivation are separable, that separating them makes approve-once coherent at every tier, and an account of what an agentic system built on that separation actually does when a model is wired into it.

11 Conclusion

Per-decision human oversight scales badly and degrades into ceremony. Rule-level oversight amortises — but only if rules generalise, and ours do not yet. What we can report is a system in which the two halves stay apart, a gate whose only question is whether a human has already ruled on this logic, and a coverage figure that is honest at 18–30 per cent along with the precise reason it is not higher.

A jury is entitled to ask whether a system that decides a fifth of its cases deterministically is worth the machinery. Our answer is that the fifth it decides, it decides the same way every time, with a derivation attached and a named person behind the logic — and that the path from a fifth to a majority runs through a mechanism we can describe, have not built, and can see the shape of in our own system’s vocabulary.